

SQL ANALYTICS · BUSINESS INTELLIGENCE

E-Commerce Data Analytics & Business Intelligence

360-Degree Retail Performance Analysis

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Project Repository: [ECommerce-Universal-SQL-Analytics](#)

Tech: SQL · CTEs · Window Functions

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Scope note. This report documents the analytical framework, methodology and findings of the SQL project. Findings stated here reflect the project's analysis outputs; figures are described qualitatively and are reproducible from the project repository.

01 Executive Summary

This report details an end-to-end SQL-based Exploratory Data Analysis (EDA) and advanced business intelligence extraction for an e-commerce retail database. Designed to transform raw transactional data (**fact_sales**) and dimensional data (**dim_customers**, **dim_products**) into actionable strategy, the project serves as a comprehensive toolkit for understanding retail performance across all major business vectors.

By segmenting customers, analysing product velocity and tracking time-series growth, the analysis moves beyond basic data extraction to provide direct strategic value to marketing, supply chain and executive leadership teams.

3 Source tables	6 Business objectives	2 Analysis phases	3 Customer cohorts
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Headline Findings

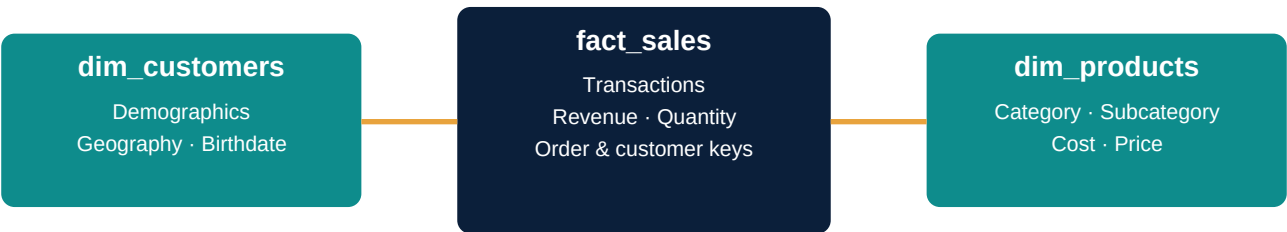
- **Top markets:** the United States, Australia and the United Kingdom lead in revenue and order volume.
- **Dominant categories:** Components and Bikes account for the largest share of sales.
- **Customer profile:** married males drove the highest global sales volume.
- **Universal best-sellers:** Patch Kit/8 Patches and Water Bottle - 30 oz. rank among the top-selling products in every country analysed.



Figure 1. Analytical workflow from raw data to segmented insight.

02 Data Architecture

The analysis runs on a star-schema retail database. A central fact table records each sales transaction, while two dimension tables describe the customers who buy and the products they buy.



Star schema: one fact table joined to two conformed dimensions

Table	Type	Role in the analysis
fact_sales	Fact	Transactional records used for revenue, volume, order and customer-activity metrics.
dim_customers	Dimension	Demographic and geographic attributes: age (from birthdate), gender, marital status, country.
dim_products	Dimension	Product category, subcategory, cost and pricing attributes used for ranking and tiering.

03 Core Business Requirements

The analytical framework was built to fulfil six strategic objectives:

#	Objective	What it delivers
1	Data Integrity	Identifies and cleans missing or anomalous data within product categories and cost structures to ensure reporting accuracy.
2	Executive KPI Tracking	A unified, high-level summary of Total Revenue, Volume, Active Customers and Average Order Value.
3	Demographic Profiling	Most profitable customer segments by age, gender, marital status and geographic footprint.
4	Product Performance	Ranks products and categories by sales volume and revenue contribution to surface high-velocity items and underperforming stock.
5	Trend & Time-Series	Year-over-Year performance, seasonality and moving averages to reveal true sales momentum.
6	Behavioural Segmentation	Product pricing tiers and customer value groups (VIP, Regular, New) from lifetime spend and account longevity.

04 Exploratory Data Analysis

The initial phase focused on understanding the data architecture, identifying anomalies and establishing baseline metrics across all dimensions.

4.1 Data Cleaning & Imputation

- Standardised empty-string values in **dim_products** by assigning consistent labels to missing categories and subcategories.
- Imputed zero-cost anomalies (specifically within *road frames*) using the historical average cost of that product category.

4.2 Customer Demographics

Profiled the customer base by unique counts, geographic distribution, marital status and dynamic age ranges derived from the minimum and maximum birthdates.

4.3 Baseline KPIs

KPI	Definition (source: fact_sales)
Total Revenue	Sum of sales amount across all transactions
Total Items Sold	Sum of quantity sold
Average Selling Price	Revenue per unit sold
Total Orders	Count of distinct orders
Active Customers	Count of distinct customers with at least one order

4.4 Magnitude Analysis

Dimension	Finding
Regions	United States, Australia and the United Kingdom are the top-performing regions by revenue and orders.
Categories	Components and Bikes are the dominant product categories.
Gender & marital status	Married males drove the highest global sales volume.

05 Advanced Analysis & Strategic Segmentation

The second phase moved from fundamental aggregations to complex business logic, using SQL Common Table Expressions (CTEs) and window functions.

5.1 Ranking & Cross-Regional Product Velocity

RANK() and **DENSE_RANK()** isolated the Top 3 revenue-generating countries and the Top 5 most frequently sold products in each. Items such as *Patch Kit/8 Patches* and *Water Bottle - 30 oz.* were dominant across all geographic borders.

5.2 Time-Series & Seasonality

Sales, order quantities and active customers were grouped by year and month to map seasonal buying patterns and the overall growth trajectory.

5.3 Cumulative Metrics & Moving Averages

WITH ROLLUP and window framing (**ROWS BETWEEN 2 PRECEDING AND 2 FOLLOWING**) produced running sales totals and centred moving averages for product prices, smoothing short-term market volatility.

5.4 Year-over-Year Performance Tracking

Comparative flags were engineered using **LAG()** by comparing current-year product sales against historical averages and the preceding year:

Flag	Comparison basis
Above Average / Below Average	Current-year sales vs. the product's historical average
Rise / Fall	Current-year sales vs. the preceding year (LAG)

5.5 Behavioural Data Segmentation

Inventory stratification. The product catalogue was categorised into defined cost tiers (for example *Below 249*, *250 - 500* and *Premium*) using CASE statements to analyse inventory distribution and pricing strategy.

Customer Lifetime Value (CLV). Customers were segmented into actionable cohorts under strict business rules:

Cohort	Business rule
VIP	More than 12 months of tenure AND over €5,000 in lifetime spending
Regular	Established customers who do not meet the VIP threshold
New	Customers with limited account tenure

Note: the VIP rule is as defined in the project; exact Regular and New boundaries are defined in the repository queries.

Representative Query Patterns

Illustrative examples of the techniques used (table and column names simplified).

```
-- Top products per country (ranking)
WITH ranked AS (
  SELECT country, product_name, SUM(quantity) AS units,
         DENSE_RANK() OVER (PARTITION BY country
                             ORDER BY SUM(quantity) DESC) AS rnk
  FROM   sales_view
  GROUP BY country, product_name)
SELECT * FROM ranked WHERE rnk <= 5;

-- Year-over-year change (LAG)
SELECT product_name, order_year, yearly_sales,
       yearly_sales - LAG(yearly_sales)
         OVER (PARTITION BY product_name ORDER BY order_year) AS yoy_diff
FROM   yearly_product_sales;
```


06 Technical Toolkit

Technique	Application in this project
CTEs	Structure multi-step logic into readable, reusable query blocks
RANK() / DENSE_RANK()	Top-N countries and products; tie-aware ranking
LAG()	Year-over-year comparison and rise/fall flags
Window framing	Centred moving averages (2 preceding to 2 following)
WITH ROLLUP	Subtotals and cumulative totals
CASE statements	Pricing tiers and customer cohort classification
Aggregations & date functions	KPIs, age ranges, year/month grouping, tenure
Data imputation	Category-average cost replacement; label standardisation

07 Business Impact by Stakeholder

Stakeholder	Value delivered
Marketing	Target high-value segments (VIP cohort, top demographics and regions); tailor retention for Regular and onboarding for New customers.
Supply chain	Prioritise stock for high-velocity items and review underperforming products using ranking and YoY flags.
Executive leadership	One consolidated KPI view plus growth momentum, seasonality and regional contribution for planning.

08 Conclusion

This project demonstrates a full analytical lifecycle in pure SQL: cleaning and validating data, defining executive KPIs, profiling customers, ranking products, tracking time-series momentum and segmenting customers and inventory into decision-ready groups.

The resulting toolkit gives marketing, supply chain and leadership teams a shared, reproducible view of retail performance, and can be extended with additional dimensions or connected to a BI layer for ongoing reporting.

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